

Introduction

Traditional electric wheelchairs heavily rely on manual input. For the millions of people affected by neurodegenerative conditions such as ALS, joysticks are ineffective mechanisms for control. Our project removes the need for physical inputs, allowing users to navigate the world using neural signals.

Current interfaces, such as SSVEP systems, require persistent cognitive focus and offer purely segmented control. This puts high demands on users who may struggle with precise commands.¹

We propose a multimodal BCI wheelchair that simplifies navigation by fusing EEG motor imagery, LiDAR mapping, and eye tracking.

Hardware

EEG

- **Wearable Sensing DSI-7:** 7 channel wireless EEG, which specializes in removing movement artifacts.

Wheelchair Model

- **Platform:** Mecanum wheel RC Car was chosen for the platform, serving as a hardware-compatible substitute for a full-sized wheelchair prototype.
- **PID Controller:** A Raspberry Pico takes the ticks from the encoder motor and runs a local PID controller that calculates how far we are from the target and controls the amount of power needed to smoothly move the motors to get to the destination.
- **Motor Control:** An Adafruit Motor HAT containing H-bridge logic, allowing for full rotational control of the motors.

LiDAR

- **Virtual Representation:** Built from Slamtec RPLiDAR A1 to reconstruct robot's environment using Python. Each point in space is described by an (r, θ).
- **Navigation:** Points are converted to (x,y) distances, then sent as commands to RC car.

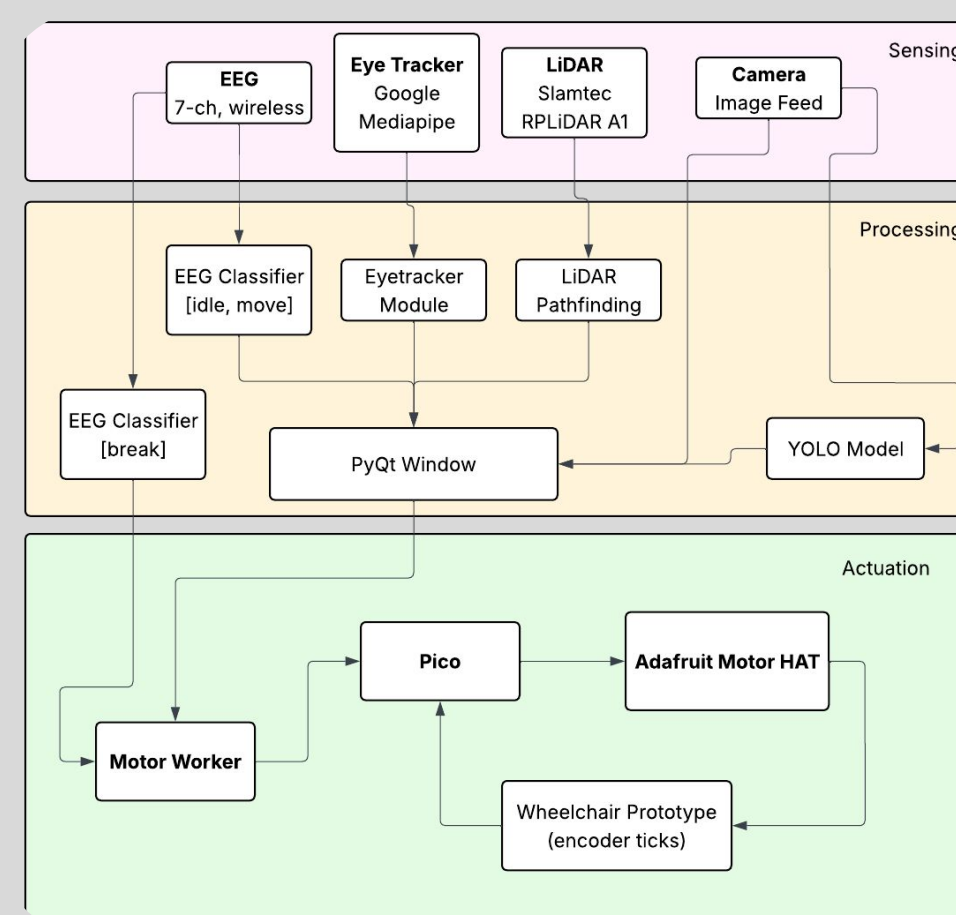
Eye Tracking

- **Face Landmarking:** Accomplished through Google's Mediapipe model.
- **Point Selection:** User gaze chooses destination point.

Visual Model

- **Safety:** Ultralytics YOLOv8 is used to detect obstructing objects in the field.

Prototype

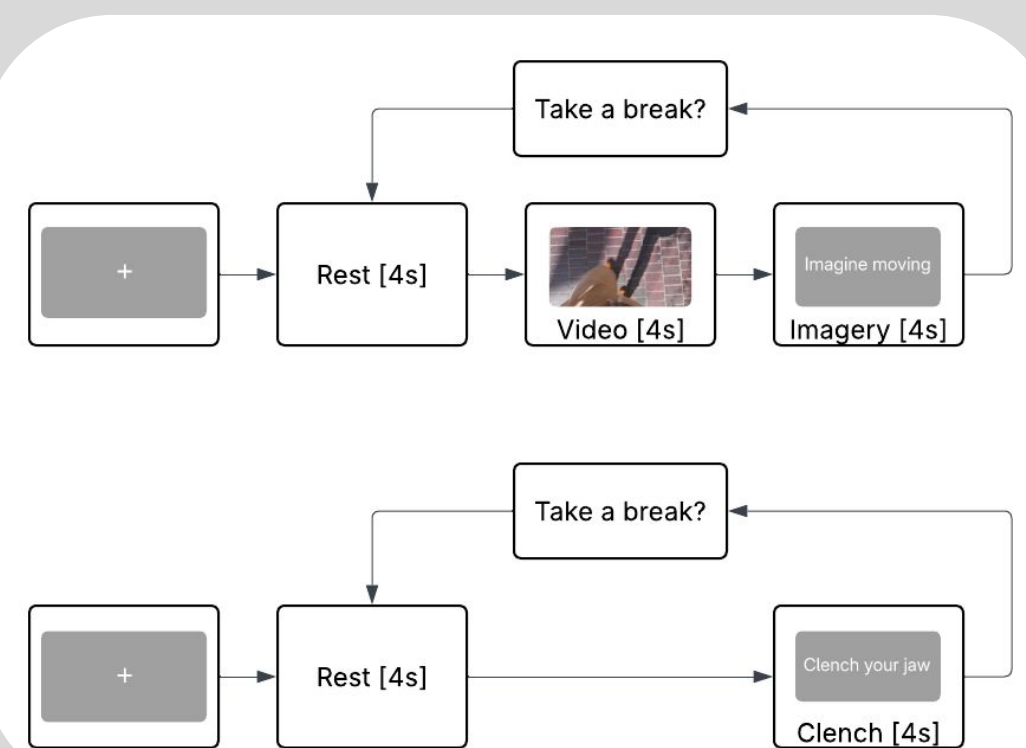


Stimulus Design

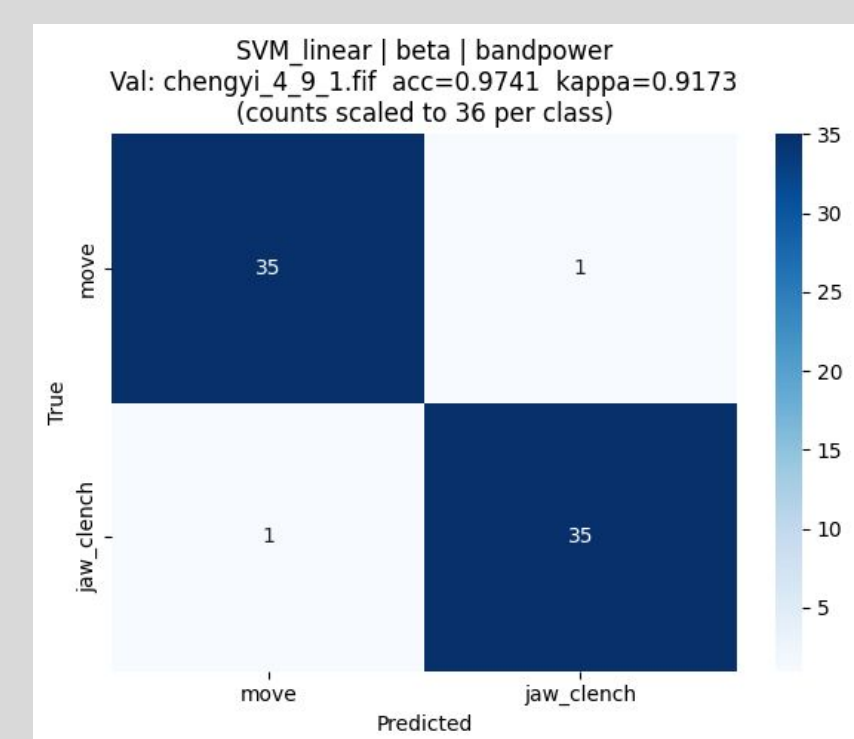
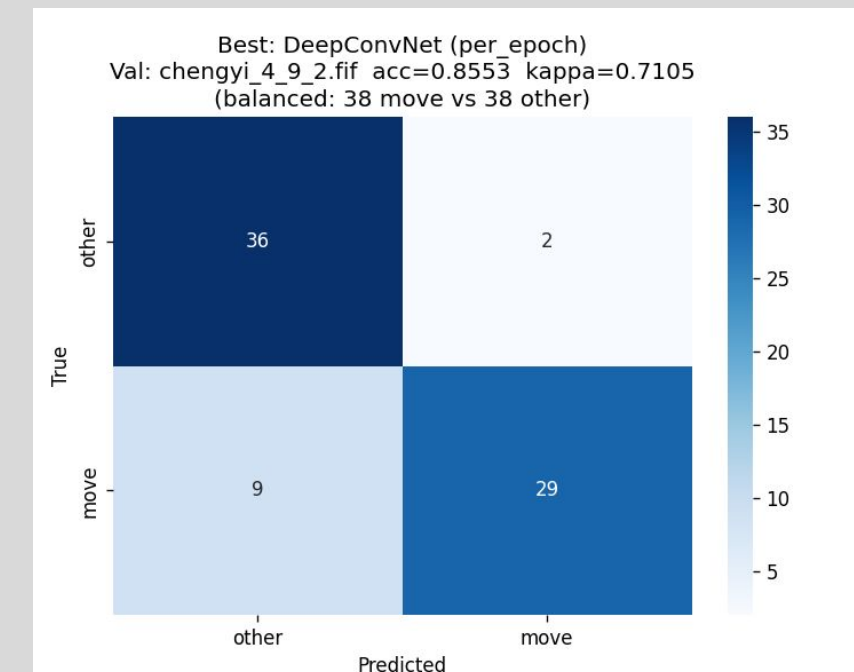
- **Goal:** Detect user intention to move and jaw movement.²

Motor Imagery (EEG)	Jaw Clench (EMG)
Go cue: user prompted to start signal	Go cue: user prompted to start signal
Fixation on screen: 3s	Fixation on screen: 3s
Ready cue + Reference Video: 4 + 4s	Ready cue: 4s
Imagery: 4s	Jaw clench: 4s
Rest: User decides	Rest: User decides

- **Total trials:** 36 trials x 2 conditions (roughly 30 mins) per session



ML Classification



- **DeepConvNet:** CNN that performs 2D convolution on multichannel time series EEG data.³
- **Idle:** Randomly selected segments of the experiment where the user was neither doing a jaw clench or motor imagery.
- **Participant:** Models are trained to be participant-specific (n=1) to aid the model's generalization.

Next Steps

- Abstract this proof of concept to a full sized wheelchair with a person.
- Expand the system to handle larger scale environments and variable (moving) parts.
- Prioritize the needs of those who would benefit from this multimodal automated wheelchair the most.
- We hope to see this product be used in the future to help those in need.

Acknowledgements

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References

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